

Synchrony Analytics Hackathon 2026 – Case Study

Theme: Customer and Credit Card Analytics – Declining Share of Wallet

Background:

XYZ Inc. has partnered with ABC Ltd. Bank to launch a co-branded credit card that offers rewards and financing options. Rewards differ based on Prime (membership) status when customers shop on XYZ Inc.

Prime members: 5% cashback on grocery purchases, 3% cashback on electronics purchases, and 1% cashback on all other eligible purchases at XYZ Inc.

Non-Prime members: 1% cashback on all purchases at XYZ Inc.

Business Problem:

Although a large number of customers hold the XYZ – ABC Co-branded credit card, ABC Ltd. leadership believes spend on XYZ Inc. using this co-branded card is declining. Instead, cardholders appear to be shifting their spend on XYZ Inc. to other payment methods (e.g., competitor cards, cash, wallets). This reduces ABC Ltd.'s Share of Wallet (SoW) on XYZ Inc.

Your Role (Analytics & Strategy team):

Assume the role of ABC Ltd. bank's Analytics & Strategy team and analyze two years of combined overall XYZ Inc. and ABC Ltd. card transaction on XYZ Inc. data to uncover customer insights and develop targeted offers/recommendations that increase ABC Ltd.'s Share of Wallet (SoW) on XYZ Inc.

Expected Outcome:

A clear set of customer segments, a model or method to identify low share-of-wallet customers, and a targeted strategy (offers/rewards) that increases usage of the XYZ – ABC co-branded credit card at XYZ Inc.

Key Considerations for Analysis:

1. Share of Wallet is defined as the proportion of a customer's total spend at XYZ Inc. that is transacted on the ABC Ltd. co-branded card.

$$\text{SoW} = (\text{Spend on ABC Ltd. card at XYZ Inc.}) / (\text{Total spend at XYZ Inc.})$$

2. SoW should be computed only for the period during which the customer held an active/open ABC Ltd. card.

3. The XYZ-ABC co-branded card currently offers the most competitive reward rates on XYZ Inc. for Grocery and Electronics purchases. No competing credit or debit card in the market matches these rates on the two categories.
4. Rewards earned in a given month are credited as a statement adjustment in the following month's billing cycle.
5. There are no joining or annual fees for XYZ-ABC co-branded card.
6. SoW must be calculated on net sales (purchases minus returns), not on gross sales.
7. Number of transactions to be considered as zero for Returns in the data.
8. XYZ Inc.'s fiscal year runs August through July.
9. Shopping categories can be determined from the Category Code table using *Category_Code* as the primary key.
10. Transaction payment methods can be identified from the Payment Code table using *Payment_Code* as the primary key.
11. Clearly state all assumptions made during the analysis.

Note: Please submit your findings/analysis in **PPT or PDF** format, along with the **code packaged as a ZIP file**, to analyticshiring@syf.com by **12:00 PM IST on 14th August**. All submission files must be named using only the student roll number.

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